BT PropNet — a commercial property trading service for the Internet

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The UK commercial property market has a large number of stakeholders which can hinder identification and analysis of key information. By using the latest Internet technology, this information can be made available to the desktop to significantly enhance the property trading business. This paper provides an overview of the recently launched BT PropNet, a trial property trading service for the UK commercial market, and describes its system infrastructure.

1. Introduction

The UK commercial property market is largely conventional in nature, where the successful matching of buyer to vendor can be a tricky process involving many commercial contacts and a variety of lengthy information searches. Key information is distributed among a large number of stakeholders, hindering both identification and suitability analysis of available properties. The Internet has the potential to enhance and simplify the property trading process substantially. Latest Internet technology allows instant, accurate matching of buyers to vendors across large property portfolios, and rapid provision of a wide range of property-related information to the desktop. Advanced features such as automated electronic agents maximise the chance of a sale, whereas new Internet visualisation technology such as interactive 'virtual reality' property tours significantly enhance publicity material.

BT has now launched on trial 'BT PropNet' [1], its own property trading service for the UK commercial market, operating over the Internet, which is already one of the most advanced services in its field.

This paper describes the PropNet service, including service operation, competitive edge features, and an insight into the system infrastructure.

2. PropNet overview

The service has two main components. Firstly, a publicly available Internet service for the property seeker, providing a range of advanced market matching and property information facilities. Secondly a secure, private service for the content provider enabling effective load and update of property details and related information. The content provider is usually a medium-large property marketing organisation or property managing outfit, such as a commercial property estate agent or a large company corporate property group. The service is outlined in Fig 1.

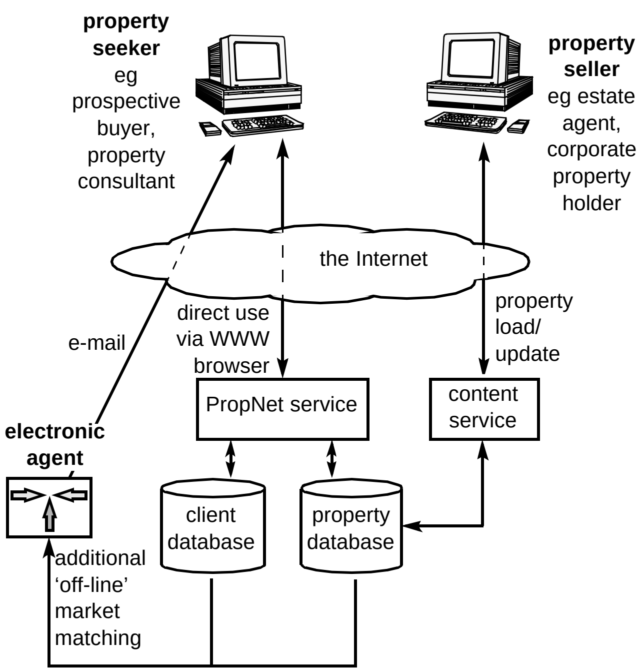

Fig 1 PropNet service overview.

A property seeker can access the public PropNet service via the Internet, needing only a standard Internet connection and a World Wide Web (WWW) browser, for example Netscape 2.0+ or Microsoft Internet Explorer 3.0+. The public service provides the user with managed, flexible access to the 'property database' and protected access to the client database. The property database contains full details on all current property issues, where a wide range of features are available to the user for searching on availability and manipulating property-related details. The client database is used for storing personal information such as a client's property requirements or specific properties of interest, which is protected by user-name and password on a per client basis.

The content provider makes use of a simple content service to load properties into the database and to perform subsequent information updates. Content load is via a secure Internet route between the content provider and BT, or via a third party authorised to perform content load on the seller's behalf. Uniform presentation style of WWW content from all providers is achieved using content load templates, where the content loader 'fills in the blanks' with custom property details.

The electronic agent uses the client database and property database to perform extended market matching, and is described in section 4.

3. Features summary

PropNet features are focused around 'market matching', i.e. bringing property seeker and property vendor together, quickly and effectively.

On entering PropNet, the property seeker can access one of four property search engines. The primary search engine provides a rapid search for any specified property types, from high street retail outlets to out-of-town land development sites. Two further search engines provide more specialised searches for the more common retail and office categories. The fourth search engine provides a powerful, generic search working across all the most prominent UK property Internet sites, described in full in section 6.

All search parameters are kept simple and flexible and the search will be as broad or as narrow as the property seeker sees fit. For example, a location address can be specified in terms of a general geographic region selected using checkboxes, or as any part of a normal address such as town, postcode or part postcode, or street number. Some sample opening pages are shown in Fig 2.

Having invoked one of the search engines, details of properties matching the user's requirements are displayed in a tabulated form, with ten properties per page. The user can flip through these pages at will to identify properties of main interest, and for those properties progress by a hyper-link to full property details. The user can display full details in any appropriate form, from either a quick reference page, to 'glossy-brochure quality' multi-page descriptions, to in some cases full virtual reality (VR) tours using Apple QuickTime VR™ technology [2]. Property details can include descriptions, colour photographs, floor plans, rent tables, and local or wide area maps. At each stage of information presentation, a link is provided to a page containing details of the property seller, providing phone/fax details along with an e-mail option where appropriate (see Fig 3).

A news feature keeps property seekers updated on the latest property offers from the PropNet portfolio, and the latest service enhancements. In addition the user has on-line access to a variety of added value features including a personal folder used for creating a particular catalogue of favoured properties, extended market matching using an electronic agent, a VR show case, and on-line help providing tips on how to use each feature. These additional features are discussed in detail in the following sections.

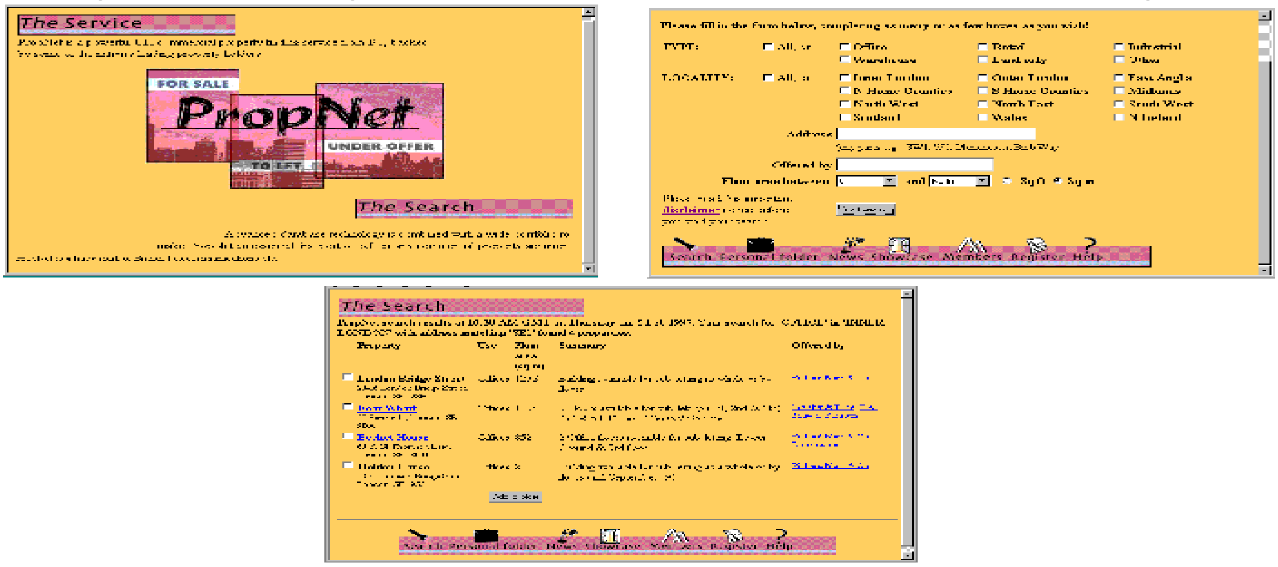

Fig 2 The PropNet front page, general property search page, and example tabulated search results.

Fig 3 Example property pages.

4. The electronic agent

The electronic agent performs extended off-line market matching, working independently of the main service. Subject to the permission of each individual property seeker, the electronic agent looks for matches of known customer requirements (stored in the client database) to all new instructions. Where new matches are found, the electronic agent prepares a custom match report for each client, e-mailing findings accordingly.

In this way a property seeker always obtains the most up-to-date information on possible purchase issues, potentially within minutes of the property becoming available, without any manual contact from the provider. The electronic agent ensures that potential sales are not missed, where each buyer always has the full, focused list of potential properties for purchase on hand.

To ensure a balanced number of communications with each property seeker, the agent attempts to combine match reports across all the user's requirements into a single e-mail. The frequency of mailing is dependent on the time between new content loads and the number of new matches found.

Individual property seekers can control the way in which the electronic agent works on their behalf using their personal folder (see section 5). This provides a simple-to-use WWW front end for creating, editing, and deleting new property requirements.

5. Service customisation

To facilitate elegant, customised use of the public service, property seekers can create their own personal folder. This folder is used to create an individual catalogue of properties of special interest, and to store the user's current property requirements. Users access their personal folder's simply by entering the correct user name and password at the folder's front page, necessary only once per session.

The personal catalogue and requirements are accessed and modified through simple menus, as shown in Fig 4. Requirements can be recalled and run at any future date, whereby a search of the database is executed against the requirement to obtain simple, focused updates on property availability. Crucially, to save daily re-visiting for new updates, these requirements can be passed to the 'electronic agent' for automatic, customised market matching, described in full in section 4.

6. Achieving critical content mass — 'the power searcher'

A medium-large sized commercial estate agency will have around one hundred properties for sale at any one time. The largest UK corporate property holders have disposal portfolios of around one hundred to two hundred properties. Thus, even with a number of medium-large customers, the property service will only span a portfolio of say five hundred to a thousand properties. Since the technology is new, a critical mass is required to raise enough market interest to make a viable service.

This critical mass must greatly exceed the typical portfolio of any one agent, or partnership of agents, or the property seeker may gain little over and above a visit to the local agent. For maximum impact, PropNet requires service partners in order to boost the searchable portfolio.

Historically, achieving such partnerships would have required a significant investment of management time, plus technically challenging integration of comparable databases across different private networks. With the public Internet, such investment is not required — an automated PropNet intelligent agent can simply visit other leading Internet property listings via their public interfaces and index content. PropNet itself can then search across a quantum increase in property stock, with turn around speeds often faster than a visit to any single third party site, due to the incorporation of very high-end indexing systems.

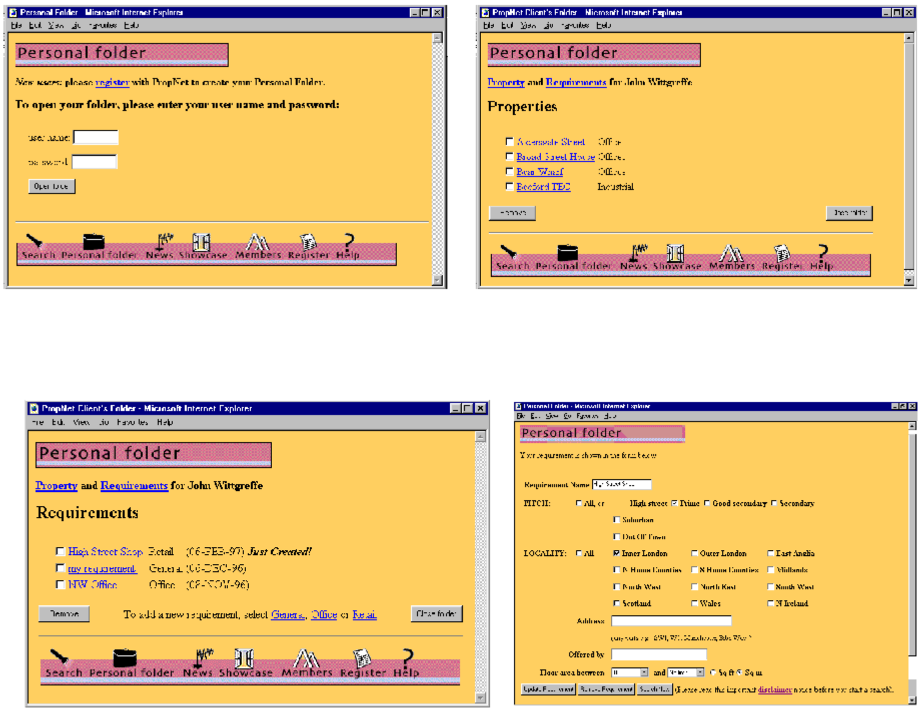

Fig 4 Property catalogue and requirements pages.

At present there are three third party UK property sites with significant content [3—5], which together with PropNet boost the searchable stock to over 6000 properties, approaching a critical mass figure.

The power search acts in the same manner as the well-established Internet search engines — properties are only obtained from free, publicly available Internet listings. When third party property listings are referenced by PropNet, the descriptions are short extracts, presented along with full details of the originating third party site. To obtain further details for any property the user must visit the third party site using the appropriate hyperlink. The third party seller thus obtains greater advertising coverage, with increased chance of a sale. Figure 5 depicts the power search architecture.

It should be noted that, at the time of writing, the power searcher is not yet available from the public PropNet service and is awaiting launch clearance.

7. Virtual reality property tours

The Internet offers increased potential for ‘experience-enhancing’ marketing of more prestigious properties. In particular, AppleQuickTime VR technology [2] and similar technologies offer the chance to walk through properties using real photographic imagery, look around in 360 °, look up, look down, zoom up to (and even pick up) objects of interest. Building tours are constructed from 360° high-resolution panoramic photographs for various scenes both inside and outside, which are then chained together into a navigable tour (see Fig 6). Music and voice commentary can be included as appropriate, and virtual reality modelling language (VRML) models of interesting features can be superimposed for more detailed study. File sizes for each panoramic scene can be as small as 15K bytes, although a full-screen, high-quality image requires at least 200K bytes. PropNet presents a special ‘VR show case’ where users can tour some of PropNet’s most prestigious properties. A useful comparison of the current leading ‘panoramic’ VR technologies is available from ‘The Network Group’ Internet site [6].

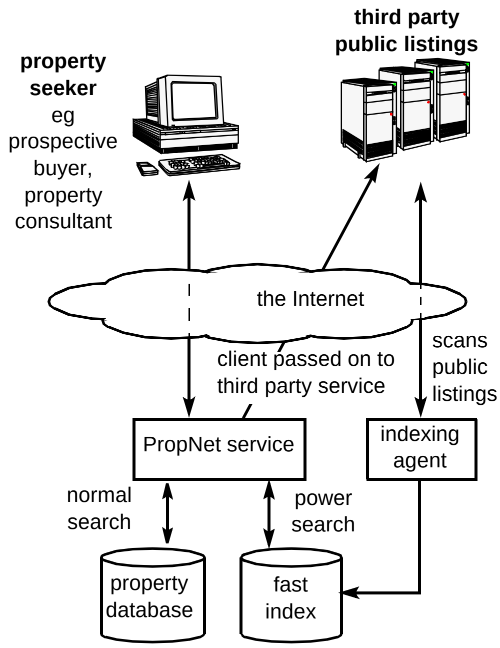

Fig 5 ‘The power search’ architecture.

Fig 6 Frames from a VR property tour.

8. Importance of a clear, fast interface

New users have a low level of computer expertise and use the site on a weekly, rather than daily, basis. The design emphasis is therefore on ease of use, learning and recall. Achieving the objective of 'best-of-breed' usability positively differentiates PropNet from other competitor property sites.

For a successful, high-impact service, the temptation is to adorn the service pages with elegant imagery designed to impress. However, such measures can have a detrimental effect. For regular users of the service, the key is speed and accuracy, over typically 14.4 kbit/s or 28.8 kbit/s modems and client computing of average power, where large file sizes are a substantial inhibiting factor.

The PropNet service makes use of clear, self-explanatory banners with brief, straightforward service pages, which are low in download size, yet pleasing to the eye. Every facility is only a few mouse clicks away from any point in the service, facilitated by careful page design to a consistent page layout template, and a persistent features menu at the base of each page. To assist new users, context-sensitive help is available at a single mouse click from any point in the service, where tips on usage are tuned to the activity in question. The download file size of each page is kept to a minimum using 'miniature' versions of photographs, allowing the user to decide whether or not to proceed to download large images based on their own assessment of the miniature. Crucially, every effort has been made to tune the property database for performance, such that even the most complex property searches are turned around in only a few seconds.

9. Charging model

At present the model is comparable to an advertising model, where all features of the public PropNet service are free to any property seeker, and the content provider is charged to load properties on a per property basis. For a successful, profit-making service, the charge is very low compared to the traditional costs of property marketing and selling, and hence attractive to property sellers. However, the technology is only in its infancy in terms of market impact, as knowledge and commitment continue to spread through the property community, and a significant portion of property individuals are yet to become computer literate.

At present significant penetration in the UK is estimated to take a further three to five years. Assuming significant uptake on these time-scales, the potential rewards are substantial, and BT is ideally placed to act as an 'electronic middle man' in providing service, with a combination of intelligent technology and communications infrastructure.

As the user-base expands, scope will appear for higher added-value subscription services, providing a whole package of property-related information on line for a small fee.

Examples of service for which property seekers and property executives might be willing to pay are historical property deals databases, rates information, individual property or site histories, current planning permissions, etc. This could be achieved by working in partnership with well-established property consulting services to ensure provision of content with a high level of interest within the community. Two UK Internet services are already experimenting with such subscription-based activity [4, 7].

10. Conclusion

The UK commercial property market relies upon the successful matching of buyer to vendor which can involve many contacts and lengthy information searches. The latest Internet technology allows this key information to be identified, analysed and presented to the desktop, using advanced features such as automated electronic agents to maximise the chance of a sale, and Internet visualisation technology to provide virtual reality property tours. BT's recently launched trial service, BT PropNet, is already one of the most advanced services in its field, providing a property trading service for the UK commercial market. This paper has given an overview of the PropNet service and described the system's communications infrastructure and intelligent technology.

Acknowledgements

Thanks to William Bland, Maije Tatiegi, Habib Zaiour, and David Skingsley for development expertise; Mark Pattinson and Farilie Gibson for user interface design; Steve Sim and Steve Elliott for system design consultancy; Ric Parker, Julian Bye, Wayne Cunningham and Paul Woods for initial WWW content editing; John Flatman of the Boots Group plc for requirements consultancy, and the team at 'The Network Group Ltd' for preparation of panormic VR material.

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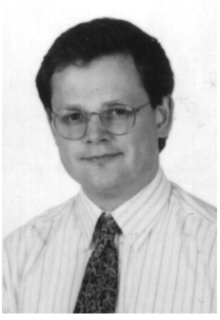

John Wittgreffe joined BT in 1989 after graduating at the University of York with a first class honours degree in Physics, initially working on research into opto-electronic materials and devices. After a move into the high demand information technology area in 1992, he has led the development of a number of database and visualisation systems, including, more recently, Internet services such as Propnet and the Microsoft/BT/BBC/Hansard joint venture the 'Schools Election '97 Polling System'. He is a member of the Institute of Physics.

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Kim Fisher qualified as a product designer in 1975, with a BA honours degree in Industrial Design Engineering. He then worked for Russell Hobbs moving to Gillette UK Research Laboratories in 1978 and then Pye Telecom in 1979 (later Philips Radio Communication Systems). In 1985, he joined BT Laboratories to set up the industrial design unit, becoming industrial manager for the BT Products Division in 1989. In April 1990, he joined the new advanced concepts unit in the Systems Research division and is currently working in the Centre for Human Communications at BT Laboratories.

Scott McRae graduated from the University of Glasgow in 1989 with an Honours BSc in Computing Science. He joined the Human Factors unit at BT Laboratories to work on workflow analysis, design and prototyping of customer, service and network management GUIs, including national and international network traffic management systems, operator workstations, ServiceView manager, and ServiceView GUI prototype for Topgear (now Tele-marketing services). In 1994 he moved into the on-line and multimedia area, leading navigation research projects and requirements capture, prototyping and bid creation for WWW applications, including PropNet, TranSend, SCARS and Flexible Work Consultant. He is responsible for the BT On-Line Usability Guideline.